

Interpolation, Curve Fitting and Calibration

Engineering Computation with Python — Chapter 12

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Two Questions About Discrete Data

Measurements are taken at discrete points, but engineering questions rarely land exactly on one of them. This chapter covers two complementary answers.

Interpolation

Estimates values **between** measured points. Passes exactly through every known point.

Curve Fitting

Finds the single model that best describes a whole **noisy** dataset. Does not need to pass through any particular point.

The choice that matters

Clean, exact data → interpolate. Noisy, scattered measurements → fit. Curve fitting is the basis of **sensor calibration**.

Interpolation	Curve Fitting
Computes missing values between two known points. Original data points are preserved — the interpolant passes exactly through them.	Converts data points into a mathematical equation. The fitted curve does not need to pass through the original data points.

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One-dimensional interpolation in SciPy follows a consistent pattern: build an interpolating function once, then call it at any query points.

```
from scipy.interpolate import interp1d

# Define the interpolation function
f = interp1d(x, v, kind=method)

# Interpolated values at xq
vq = f(xq)
```

`numpy.interp()` does the same job for the linear case alone, with no object to build first.

Parameter	Description
x	Sample points (array-like).
v	Corresponding values at the sample points.
xq	Points at which to evaluate the interpolation.
method	'linear', 'nearest', 'cubic', etc.
vq	The interpolated values at xq.

Every linear-interpolation call, under the hood, evaluates one formula. Given two known points (x_0, y_0) and (x_1, y_1) , the value y at a query point x with $x_0 \leq x \leq x_1$ is

$$y = y_0 + (x - x_0) \frac{y_1 - y_0}{x_1 - x_0}.$$

What this is, geometrically

A straight line fitted **between the two known points**, evaluated at x . This is exactly what `numpy.interp()` and `interp1d(..., kind='linear')` compute.

Method	Description	Min. Points
nearest	Steps to the value of the nearest data point.	2
linear	Straight lines connect the points.	2
cubic	Piecewise cubic spline.	4

Choosing a method is choosing an assumption

nearest assumes the signal holds its value until the next sample; linear assumes constant rate of change; cubic assumes smoothness. None of these is automatically correct — the right choice depends on the physics being measured.

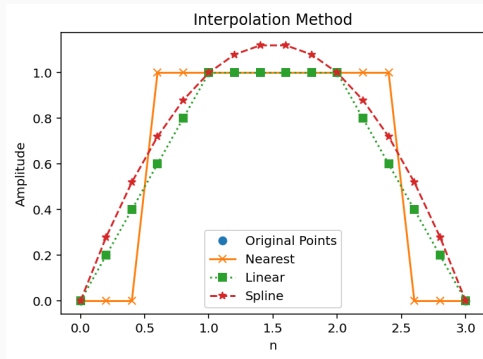
Worked Example: Comparing Interpolation Methods

The same four points, interpolated three different ways:

```
x = np.array([0, 1, 1, 0])
n1 = np.arange(0, 4)
n2 = np.arange(0, 3.2, 0.2)

xnearest = interp1d(n1, x, kind='nearest')
xlinear = interp1d(n1, x, kind='linear')
xspline = interp1d(n1, x, kind='cubic')

plt.plot(n1, x, 'o', label='Original Points')
plt.plot(n2, xnearest(n2), '-x', label='Nearest')
plt.plot(n2, xlinear(n2), ':s', label='Linear')
plt.plot(n2, xspline(n2), '--*', label='Spline')
```



Nearest produces a step-like response. **Linear** produces straight-line segments with visible corners. **Spline** produces a smooth curve — but notice it **overshoots** above 1 and below 0, values the original data never contained.

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The API describes air quality from readings at monitoring stations, adopted by Malaysia's Department of Environment in 1996 (based on the US-EPA Pollutant Standards Index).

By the way: understanding the API scale

API Range	Health Status
0–50	Good
51–100	Moderate
101–200	Unhealthy
201–300	Very Unhealthy
301+	Hazardous

During a haze crisis, API readings are recorded at five stations along the route from Kulai to Johor Bahru:

Station	Kulai	Senai	Skudai	Perling	JB
Distance (km)	0	6	15	23	30
API value	100	180	199	280	310

The question

What is the API at a location with **no sensor** — say, kilometre 10? Assume the API varies linearly between neighbouring stations.

A helper normalizes the input shape first:

```
arraykm = np.atleast_1d(km)
```

This accepts a single integer, a list, or a NumPy array uniformly — one code path, no type-checking branches.

```
def API_interpolation(km):  
    distances = np.array([0, 6, 15, 23, 30])  
    api_values = np.array([100, 180, 199, 280, 310])  
    full_distances = np.arange(0, 31, 1)  
    interpolated_api = np.interp(full_distances,  
                                distances, api_values)  
  
    km = np.atleast_1d(km)  
    result = {}  
    for k in km.astype(int):  
        result[k] = round(interpolated_api[k], 2)  
    return result
```

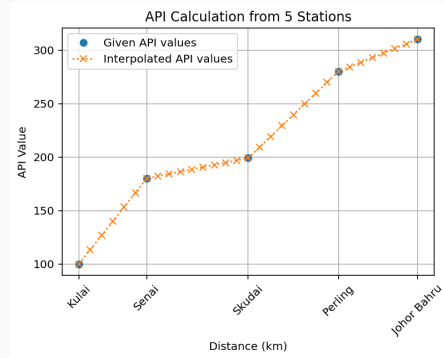
```
API_interpolation(10)           # Single integer  
API_interpolation([3, 8, 11, 26]) # List of km points  
API_interpolation(np.linspace(0, 30, 5)) # Evenly spaced points
```

API value at km 10 is 188.44

API value at km 26 is 292.86

Notice

One function, three call shapes, zero extra code — the payoff of normalizing input with `np.atleast_1d` up front.



The five known stations (circles) anchor a straight-line estimate (dotted) for every kilometre in between.

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Two broad uses recur throughout electrical engineering:

- **Evaluating unknown values** between known points — the API example just shown.
- **Increasing the detail (resolution)** of an undersampled signal.

Upsampling

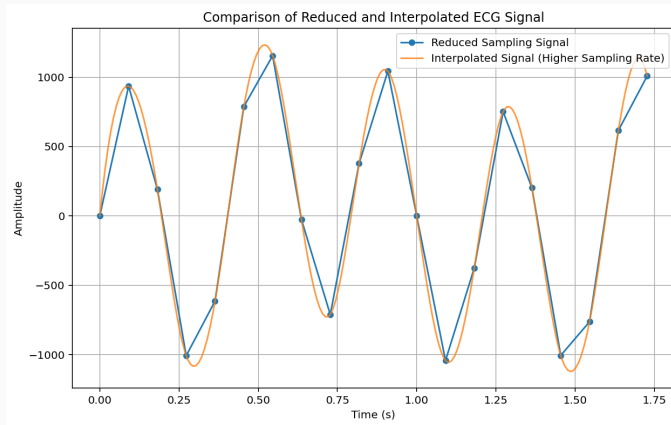
Adding points *between* existing samples by interpolation is called **upsampling**. It cannot invent information the sensor never recorded — it can only estimate plausible values consistent with the samples that exist.

A wireless ECG device under-samples to save energy. Spline interpolation adds points between the original samples:

```
reduced_ecg_data = pd.read_csv('reduced_ecg_signal.csv')
time_reduced = reduced_ecg_data['Time(s)'].values
amplitude_reduced = reduced_ecg_data['Amplitude'].values

spline_interpolation = interp1d(time_reduced,
                                amplitude_reduced, kind='cubic')

new_time_high = np.linspace(time_reduced.min(),
                             time_reduced.max(),
                             len(time_reduced) * 10)
new_amplitude_high = spline_interpolation(new_time_high)
```



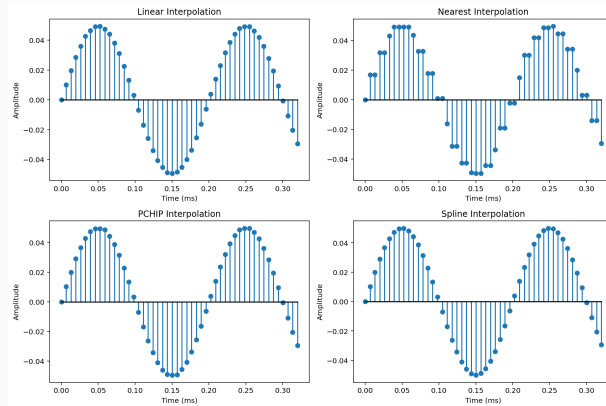
The original 100-point recording (markers) versus the $10\times$ -upsampled spline interpolation (smooth line) – the same heartbeat, described with far more detail between samples.

A digital-to-analog converter (DAC) – the final step in playing digital audio – benefits from interpolation *before* conversion, simplifying the lowpass filter that follows:

Audio Signal → Interpolation (upsampling) → Lowpass Filter → DAC

- **Linear**: straight lines between points.
- **Nearest**: step-like, assigns the nearest sample.
- **PCHIP**: smooth, **no overshoot**, preserves monotonicity.
- **Spline**: smooth, continuous derivatives, **can** overshoot.

Comparing Four Methods on a DAC Signal



Linear and spline look visually similar; nearest shows the characteristic steps; PCHIP tracks the data closely without spline's overshoot risk.

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Given scattered, noisy measurements, what single line $y = mx + b$ best describes the trend?

`np.polyfit()` answers this directly:

```
x = np.array([0, 1, 2, 3, 4, 5])
y = np.array([1.1, 2.9, 5.2, 6.8, 9.3, 11.0])

m, b = np.polyfit(x, y, 1)
print(f"Best-fit line: y = {m:.2f}x + {b:.2f}")

y_fit = np.polyval([m, b], x)
```

Best-fit line: $y = 1.99x + 1.03$

Unlike interpolation, this line does **not** pass through any of the original noisy points exactly – it summarizes the trend.

The same `polyfit/polyval` pair scales to any degree:

```
p = np.polyfit(x, y, 3) # Fit a 3rd-degree polynomial  
y_fit = np.polyval(p, x) # Evaluate it
```

Why model with a polynomial at all?

Automating a process, analyzing the relationship between variables, and identifying trends — e.g. modelling yearly climate data or internet-usage growth.

Worked Example: Sensor Calibration

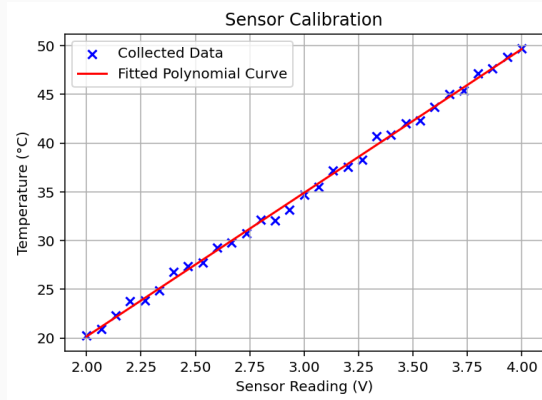
A temperature sensor reads in volts over 20–50°C. Find the function converting volts to temperature:

```
df = pd.read_excel('sensor_calibration.xlsx')
V, T = df['V'].values, df['T'].values

p = np.polyfit(V, T, 1)    # linear fit
s_t = np.polyval(p, V)
print(f'T(V) = {p[0]:.2f}V + {p[1]:.2f}')
```

$$T(V) = 14.74V + -9.33$$

This is sensor calibration: one `polyfit()` call turns a raw voltage reading into a trustworthy temperature.



Collected data (crosses) with the fitted linear calibration curve $T(V) = 14.74V - 9.33$ (line).

Six months of internet usage (hours) for five first-time users. Task: find the trend and predict months 7–9.

Person	M1	M2	M3	M4	M5	M6
1	7	11	23	59	120	180
2	3	8	20	48	88	140
3	4	11	21	58	128	195

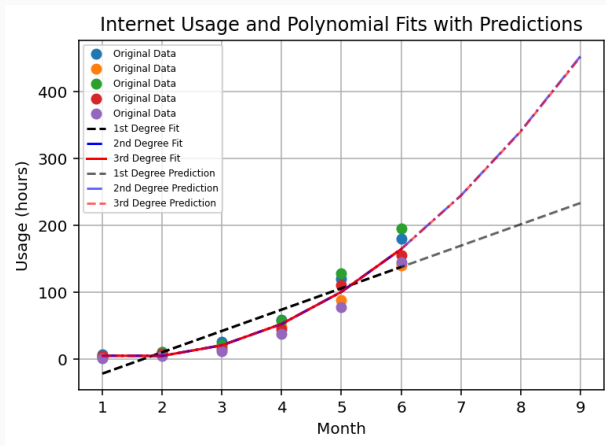
Fit 1st-, 2nd-, and 3rd-degree polynomials and compare their predictions.

```
months = np.arange(1, 7)
month_matrix = np.tile(months, (data.shape[0], 1))

p1 = np.polyfit(month_matrix.flatten(), data.flatten(), 1)
p2 = np.polyfit(month_matrix.flatten(), data.flatten(), 2)
p3 = np.polyfit(month_matrix.flatten(), data.flatten(), 3)

months_predict = np.array([7, 8, 9])
usage_1 = np.polyval(p1, months_predict)
usage_2 = np.polyval(p2, months_predict)
usage_3 = np.polyval(p3, months_predict)
```

1st order: [169.7 201.6 233.4] 2nd order: [244.5 340.5 452.4] 3rd order: [244.3 340.0 451.4]



The 1st-degree (linear) fit substantially **underestimates** the accelerating growth. The 2nd- and 3rd-degree fits track it closely and agree with each other.

Worked Example: Water Usage and R^2

A farmer wants an automatic irrigation formula from one harvest cycle's water-usage data. Which polynomial degree is the right choice?

R^2 : how much variation the fit explains

$$R^2 = 1 - \frac{\sum_i (d_i - f_i)^2}{\sum_i d_i^2}$$

d_i = measured data, f_i = fitted data. Closer to 1 is better – but **not the whole story**.

```
N = 5
water_fit = [np.polyval(np.polyfit(day, water, n), day)
              for n in range(1, N + 1)]

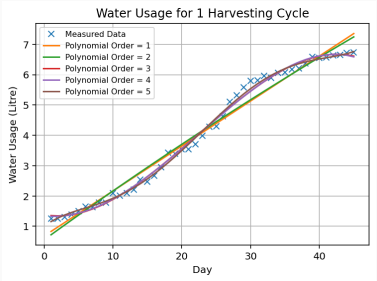
R2 = [1 - np.sum((fit - water)**2) / np.sum((water - np.mean(water))**2)
      for fit in water_fit]
```

Order 1 = 0.9743 Order 2 = 0.9750 Order 3 = 0.9947 Order 4 = 0.9947 Order 5 = 0.9961

Diminishing returns

R^2 jumps from 0.975 to 0.995 going from order 2 to 3, then barely moves beyond that. Order 3 is the sensible choice: good accuracy, no unnecessary complexity.

Why Not Always Pick the Highest R^2 ?



Overfitting

A polynomial with enough degrees can track *every wiggle* in noisy data, including the noise itself – it then describes only **this particular sample**, and predicts poorly on new data.

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The nominal time constant of the bench RC circuit is $\tau = RC = 0.47 \text{ s}$ — but that assumes the resistor and capacitor are exactly their marked values. Real components carry tolerances.

The honest answer comes from the data

What is the **measured** τ , from the actual voltage recording? This is curve fitting applied to the chapter's own running example.

$v(t) = V_0 e^{-t/\tau}$ is not a straight line, and `np.polyfit()` fits polynomials. Taking logarithms of both sides:

$$\ln v = \ln V_0 - \frac{t}{\tau}$$

This *is* a straight line in t

Slope = $-1/\tau$, intercept = $\ln V_0$. Fitting a degree-1 polynomial to $\ln v$ recovers **both** parameters at once.

Recovering τ from the Data

```
t = np.arange(0, 2.001, 0.02)
v = 12.0 * np.exp(-t / 0.47)  # stands in for measured voltage

slope, intercept = np.polyfit(t, np.log(v), 1)

tau_measured = -1 / slope
V0_measured = np.exp(intercept)
```

Fitted slope: -2.12766 per second

Measured tau: 0.4700 s

Measured V0: 12.0000 V

Three lines of code measure the circuit's **actual** RC product, rather than assuming the marked component values.

1. The logarithm is undefined for $v \leq 0$

Any sample that noise has driven to zero or below must be excluded *before* fitting.

2. Logarithms distort the error weighting

They compress large voltages and stretch small ones, so the tail of the decay influences the fit more than it would in a direct fit. Fine for a first estimate; for precision work, a nonlinear fit to the original exponential is the better tool.

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- **Interpolation** passes exactly through measured points; **curve fitting** finds the best model for a noisy whole dataset. Noisy data calls for fitting, not interpolation.
- `np.interp()` does linear interpolation directly; `interp1d()` adds nearest, cubic, and spline methods via `kind`.
- The method must suit the physics: nearest $\rightarrow$ steps, linear $\rightarrow$ corners, cubic/PCHIP $\rightarrow$ smooth curves (but spline can invent overshoot the data never contained).
- `np.polyfit(x, y, n)` fits a degree- n polynomial; `np.polyval()` evaluates it. Degree 1 is a straight-line trendline — the basis of sensor calibration.
- R^2 measures explained variation, but always rises with degree. A polynomial that tracks every wiggle is **overfitted** and predicts poorly.
- Not every model is a polynomial: an exponential decay becomes linear under logarithms, recovering the bench circuit's τ with the same `polyfit()` tool.

This chapter answered *what value lies between (or near)* the data you have. The next chapters build on exactly this foundation.

- Chapter 13: numerical calculus — derivatives and integrals computed directly from arrays, and polynomial models of system behaviour.
- The bench circuit's measured τ , found here by curve fitting, reappears as a check against the calculus-based methods in the next chapter.

Next: Numerical Calculus and System Models